$$\ell_t(\theta) = \sum_{k=0}^{K} \left[\ell^p(\pi_{t+k}, p_t^k) + \ell^v(z_{t+k}, v_t^k) \right] + \sum_{k=1}^{K} \ell^r(u_{t+k}, r_t^k) + c\|\theta\|_2^2$$